5.1 .

#1

(1) $(x+1)y'' + y' = 0$

$\Rightarrow y'' + \frac{1}{x+1}y' = 0$

$q(x) = \frac{1}{x+1}$ 은 $x=0$ 에서 해석적 //

$\therefore y = \sum_{n=0}^{\infty} a_n x^n$ 표현가능

.

$xy'' - y'' + y' = 0$

$\sum_{n=2}^{\infty} a_n \cdot n(n-1) \cdot x^{n-2} - \sum_{n=2}^{\infty} a_n \cdot n(n-1) \cdot x^{n-2} + \sum_{n=1}^{\infty} a_n \cdot n \cdot x^{n-1} = 0$

$\sum_{n=1}^{\infty} a_{(n+1)} \cdot n \cdot (n+1) \cdot x^{n} - \sum_{n=0}^{\infty} a_{(n+2)} (n+1)(n+2) x^{n} + \sum_{n=0}^{\infty} a_{n+1} \cdot (n+1) \cdot x^{n} = 0$

$x^n \left(a_{n+1}\, n(n+1) - a_{n+2}(n+1)(n+2) + a_{n+1}(n+1) \right) = 0$

$\therefore a_{n+1}\, n(n+1) - a_{n+2}(n+1)(n+2) + a_{n+1}(n+1) = 0$

$a_{n+2} = \frac{(n+1)^{\not}}{(n+1)^{\not}(n+2)} a_{n+1}$

a_0

a_1

$a_2 = \frac{1}{2} a_1$

$a_3 = \frac{2}{3} a_2 = \frac{2 \cdot 1}{3 \cdot 2} a_1 = \frac{1}{3} a_1$

$a_4 = \frac{3}{4} a_3 = \frac{3 \cdot 1}{4 \cdot 3} a_1 = \frac{1}{4} a_1$

$a_5 = \frac{4}{5} a_4 = \frac{1}{5} a_1$

$\vdots$

$a_n = \frac{1}{n} a_1 \quad (n \geq 1)$

$\therefore y = \sum_{n=1}^{\infty} a_n x^n + a_0$

$= a_1 \sum_{n=1}^{\infty} \frac{x^n}{n} + a_0$ //

2

(3) $y' = x - 4x + y + 1$; $y(2) = 3$

$\Rightarrow y' - y = -3x + 1$

$q(x) = -1$, $r(x) = -3x+1$ 는 $x=0$에서 해석적

$\therefore y = \sum_{n=0}^{\infty} a_n x^n$ //

$\therefore \sum_{n=1}^{\infty} a_n \cdot n \cdot x^{n-1} - \sum_{n=0}^{\infty} a_n x^n = -3x+1$

$\sum_{n=0}^{\infty} a_{n+1}\cdot(n+1)\, x^n - a_n x^n = -3x+1$

$\sum_{n=0}^{\infty} x^n \left(a_{n+1}(n+1) - a_n \right) = -3x+1$ //

$n=0 \cdots (a_1 \cdot 1 - a_0)\cdot 1 = 1 \quad \therefore a_1 = 1 + a_0$

$n=1 \cdots (a_2 \cdot 2 - a_1)\, x = -3x$

$2a_2 - a_1 = -3 \quad \therefore a_2 = \frac{1}{2}(a_1 - 3)$

$n \geq 2 \cdots a_{n+1}(n+1) - a_n = 0$

$a_{n+1} = \frac{1}{n+1} a_n$

a_0

$a_1 = 1 + a_0$

$a_2 = \frac{1}{2}(a_1 - 3) = \frac{1}{2}(a_0 - 2)$

$a_3 = \frac{1}{3} a_2 = \frac{a_0 - 2}{3 \cdot 2}$

$a_4 = \frac{1}{4} a_3 = \frac{a_0 - 2}{4 \cdot 3 \cdot 2}$

$\vdots$

$a_n = \frac{a_0 - 2}{n!}$ // $(n \geq 2)$

$\therefore y = \sum_{n=0}^{\infty} a_n x^n$

$= \sum_{n=2}^{\infty} \frac{(a_0 - 2)}{n!} x^n + a_0 + (1 + a_0)x$

$y(2) = 3$

$(a_0 - 2)(e^x - x) + x(a_0 + 1) + a_0 = y$

$3 = (a_0 - 2)(e^2 - 2) + 2(a_0 + 1) + a_0$

$= a_0 e^2 - 2a_0 + 4 - 2e^2 + 3a_0 + 2$

$= (a_0 - 2)e^2 + a_0 + 6$

$y = 2e^{x-2} - (x^2 - 2x - 1)$